

A Lower Bound at Exponent $\sqrt{3}$ for Predetermined-Stepsize Gradient Descent

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Abstract

We study the last iterate of gradient descent on convex L -smooth functions when the entire nonnegative stepsize schedule is fixed before any oracle information is observed. Our proof uses a marked-stage realization and rank-cutoff framework that also appears in Ma and Chen, who give universal lower bounds with every exponent $\gamma > \sqrt{2 + \sqrt{3}}$ [13]. Relative to their argument, our new ingredient is a direct entropy bound for the full, unrelaxed temporally ordered reciprocal product; it replaces their budget-equalization and two-matching bound. We prove that there is an absolute constant $c_{\text{lb}} > 0$ such that, for every horizon T , every predetermined nonnegative schedule, and every prescribed initialization, one can construct an instance in dimension at most $T + 1$ satisfying

$$f(x_T) - f^* \geq c_{\text{lb}} L R^2 (T + 1)^{-\sqrt{3}}.$$

The lower-bound proof is self-contained: the citations to Ma and Chen record provenance, while every shared component used below is rederived in full and no result from their paper is invoked as a premise.

Starting from exact local and terminal identities proved below, and also present in Ma–Chen, the proof retains the full temporal inverse product associated with the selected long steps. A two-state entropy inequality telescopes along the full selected path, cancels all interior entropy terms, and reduces the product estimate to a uniformly controlled pair of endpoint states. This yields a critical-density alternative at density $1/2$; cutoff propagation and a finite-cutoff argument then close at exponent $\sqrt{3}$. The gap to the silver-schedule upper exponent $\log_2(1 + \sqrt{2})$ remains open. We use ChatGPT 5.6 Pro and ChatGPT 5.6 Sol Ultra for developing part of the proof for this paper.

1 Introduction

We consider unconstrained minimization of a convex function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ whose gradient is L -Lipschitz. Gradient descent (GD) takes the form

$$x_{t+1} = x_t - \eta_t \nabla f(x_t), \quad 0 \leq t < T. \quad (1)$$

A schedule is *predetermined* when the complete vector $(\eta_0, \dots, \eta_{T-1}) \in \mathbb{R}_{\geq 0}^T$ is fixed before any function value or gradient is observed. It may depend on the announced horizon and on L , and it may be nonmonotone and contain arbitrarily large steps, but it cannot react to the realized oracle transcript.

Classical stable steps give an $O(T^{-1})$ last-iterate guarantee for smooth convex minimization. Carefully arranged long steps can do better. Although an individual long step may increase the objective, interactions among many such steps can yield a faster global rate. In particular, silver schedules attain the exponent $\log_2(1 + \sqrt{2})$, and concatenation constructions extend that guarantee to arbitrary announced horizons [1, 2]. This raises a restricted minimax question: how fast can plain GD converge when only its scalar stepsizes may be designed in advance?

A lower bound with exponent β_{lb} means that every schedule has an instance whose error is at least a constant multiple of $T^{-\beta_{\text{lb}}}$. A smaller lower-bound exponent is therefore stronger: it rules out every uniform upper rate $O(T^{-\beta_{\text{up}}})$ with $\beta_{\text{up}} > \beta_{\text{lb}}$. This reverses the visual ordering familiar from upper bounds and makes a precise minimax definition useful.

1.1 Minimax formulation and main result

We first remove the inessential scale parameters. For $d \geq 1$, a prescribed point $z \in \mathbb{R}^d$, and a dimensionless schedule $\alpha = (\alpha_0, \dots, \alpha_{T-1}) \in \mathbb{R}_{\geq 0}^T$, let $x_t^{f, \alpha, z}$ denote the iterates

$$x_0^{f, \alpha, z} = z, \quad x_{t+1}^{f, \alpha, z} = x_t^{f, \alpha, z} - \alpha_t \nabla f(x_t^{f, \alpha, z}).$$

Let $\mathcal{F}_1(\mathbb{R}^d)$ be the class of convex functions on $\mathbb{R}^d$ with 1-Lipschitz gradient, and define the worst-case normalized last-iterate error

$$\mathcal{R}_{T,d}(\alpha; z) := \sup_{\substack{f \in \mathcal{F}_1(\mathbb{R}^d), x_* \in \arg\min f \\ \|z - x_*\| = 1}} \{f(x_T^{f, \alpha, z}) - f(x_*)\}. \quad (2)$$

This value is independent of z . Indeed, for $z, z' \in \mathbb{R}^d$, the translation

$$f'(x) := f(x + z - z'), \quad x'_* := x_* - z + z'$$

is a bijection between the instances in (2) at z and at z' , and it translates the entire GD trajectory without changing any function-value gap. We may therefore fix the prescribed initialization to be the origin 0_d in each dimension.

The horizon- T minimax value is

$$\mathcal{M}_T := \inf_{\alpha \in \mathbb{R}_{\geq 0}^T} \max_{1 \leq d \leq T+1} \mathcal{R}_{T,d}(\alpha; 0_d). \quad (3)$$

Thus the schedule is chosen before the adversarial instance and dimension, while the hard instance may depend on the schedule. The normalization loses no generality: for physical parameters $L, R > 0$, set $\alpha_t = L\eta_t$ and apply the translation and scaling in Section 7.

Definition 1.1 (Horizon-dependent minimax exponent). The minimax exponent is

$$\beta_{\text{GD}} := \sup \left\{ \beta \geq 0 : \text{there is a constant } C_\beta < \infty, \text{ independent of } T, \text{ such that } \mathcal{M}_T \leq C_\beta (T+1)^{-\beta} \text{ for every } T \geq 1 \right\}. \quad (4)$$

The defining set is nonempty: the zero schedule and the smoothness inequality give $\mathcal{M}_T \leq 1/2$. The infimum in (3) is taken separately at each announced horizon. Thus β_{GD} describes horizon-dependent schedules; an anytime exponent, which would require one infinite schedule whose prefixes work simultaneously at every horizon, is a different quantity.

Theorem 1.2 (Universal $\sqrt{3}$ lower bound). *There is an absolute constant $c_{\text{lb}} > 0$ such that the following holds. For every $T \geq 1$, every $L, R > 0$, and every predetermined schedule $(\eta_t)_{t=0}^{T-1} \in \mathbb{R}_{\geq 0}^T$, there is a dimension $d \leq T+1$ such that, for every prescribed $x_{\text{init}} \in \mathbb{R}^d$, there exist a convex L -smooth function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and a point $x_* \in \arg\min f$ satisfying $\|x_{\text{init}} - x_*\| = R$ for which GD initialized at $x_0 = x_{\text{init}}$ obeys*

$$f(x_T) - f(x_*) \geq c_{\text{lb}} L R^2 (T+1)^{-\sqrt{3}}. \quad (5)$$

The function and minimizer may depend on the prescribed schedule and initialization, but the constant c_{lb} does not.

Corollary 1.3 (Minimax consequence). *For every $T \geq 1$,*

$$\mathcal{M}_T \geq c_{\text{lb}}(T+1)^{-\sqrt{3}}, \quad \beta_{\text{GD}} \leq \sqrt{3}.$$

Together with the arbitrary-horizon silver-schedule upper bound [1, 2], this gives

$$\boxed{\log_2(1 + \sqrt{2}) \leq \beta_{\text{GD}} \leq \sqrt{3}.} \quad (6)$$

Proof. Apply Theorem 1.2 with $L = R = 1$ to each $\alpha \in \mathbb{R}_{\geq 0}^T$, take the maximum over $1 \leq d \leq T+1$, and then take the infimum over α . This proves the lower bound on $\mathcal{M}_T$.

If some $\beta > \sqrt{3}$ belonged to the set in Definition 1.1, then for a constant C_β independent of T ,

$$c_{\text{lb}}(T+1)^{-\sqrt{3}} \leq \mathcal{M}_T \leq C_\beta(T+1)^{-\beta}$$

for every T , which is impossible as $T \rightarrow \infty$. Hence $\beta_{\text{GD}} \leq \sqrt{3}$. The cited arbitrary-horizon silver-schedule guarantee places $\log_2(1 + \sqrt{2})$ in the closure of the achievable upper exponents and proves the opposite inequality. This left-hand inequality is the only externally supplied implication in the corollary; it is not used for Theorem 1.2 or for $\beta_{\text{GD}} \leq \sqrt{3}$. $\square$

The exponent comparison is summarized in Table 1. For the lower-bound rows, the smaller exponent gives the stronger restriction.

Result	exponent	numerical value
Silver-schedule upper bound [1, 2]	$s = \log_2(1 + \sqrt{2})$	1.2715533...
Theorem 1.2	$\beta_\star = \sqrt{3}$	1.7320508...
Ma–Chen (2026) lower bound [13]	every $\gamma > \sqrt{2 + \sqrt{3}}$	$\gamma > 1.9318517 \dots$

Table 1: Upper and lower exponents for horizon-dependent last-iterate convergence.

1.2 Related work

Stepsize-only acceleration. Predetermined, time-varying stepsizes can accelerate plain GD even when occasional steps are large enough that a single iteration need not decrease the objective [3, 4, 1, 5]. Altschuler and Parrilo introduced the recursive silver schedule in the strongly convex setting and subsequently proved, for smooth convex optimization, the last-iterate rate $O(T^{-\log_2(1+\sqrt{2})})$ at horizons $T = 2^k - 1$ [4, 1]. Grimmer, Shu, and Wang obtained the same exponent for objective-gap and squared-gradient-norm guarantees using related long-step schedules [5]. Concatenation and composition constructions extend this exponent to every prescribed horizon [2, 6].

Lower bounds and the oracle model. General accelerated first-order methods attain the $O(T^{-2})$ rate [7], and the corresponding information-based $\Omega(T^{-2})$ lower bound is tight for the full first-order oracle class in sufficiently high dimension [8]. That general lower bound allows algorithms to combine past oracle information and therefore does not distinguish plain GD from accelerated methods. Structure-specific lower bounds for time-invariant oblivious algorithms were given by Arjevani and Shamir [9]; they do not cover arbitrary time-varying stepsize schedules. Grimmer, Shu, and Wang prove matching lower bounds within their restricted class of basic composable schedules [6].

Anytime versus prescribed-horizon schedules. An anytime guarantee uses one infinite stepsize sequence, independent of the stopping time, and must hold at every stopping time. This differs from (3), where the length- T schedule may depend on the announced horizon. Kornowski and Shamir posed whether anytime stepsize-only acceleration is possible [10]. Zhang, Lee, Du, and Chen constructed an anytime schedule with objective-gap rate [11]

$$O\left(T^{-2s/(1+s)}\right) = O(T^{-1.119\dots}), \quad s = \log_2(1 + \sqrt{2}),$$

Tsai, Fatkhullin, Zhang, and He subsequently proved that no single infinite schedule of strictly positive stepsizes can have worst-case objective-gap rate $o(T^{-4/3})$ [12]. Their quantifiers and strict-positivity assumption differ from the present horizon-dependent theorem, which permits nonnegative stepsizes.

Relation to the Ma–Chen framework. Ma and Chen prove that arbitrary predetermined nonnegative schedules admit a worst-case last-iterate lower bound of order $T^{-\gamma}$ for every $\gamma > \sqrt{2 + \sqrt{3}}$ [13]. Their proof contains counterparts of our marked-stage realization, residual-rank statistics, neighboring-cutoff recurrences, Lyapunov transport, and bounded-rank prefix estimate. We rederive each below; the citations record provenance, not logical dependence. Our distinct component is a direct two-state entropy bound for the full, unrelaxed ordered reciprocal product. It replaces their budget equalization, auxiliary terminal vertex, and matching analysis, yields the half-density certificate, and closes at $\sqrt{3}$. Separate notation clarifies the variables’ roles, not a different outer architecture.

Logical self-containedness. Theorem 1.2 and the inequality $\beta_{\text{GD}} \leq \sqrt{3}$ do not use any theorem, lemma, or calculation from Ma–Chen as a black box. Every paper-specific construction, recurrence, estimate, and case argument is proved in Sections 2–7; hence the result does not depend on the correctness of [13]. The cited silver-schedule bound is used only for the opposite comparison $\log_2(1 + \sqrt{2}) \leq \beta_{\text{GD}}$, not for the lower-bound theorem.

1.3 Technical contribution and proof roadmap

Main technical idea. Split each dimensionless step into a capped part and a nonnegative surplus, order the positive surpluses, choose a cutoff prefix, and restore it to temporal order. Proposition 2.1 realizes this selection as a stage chain whose final gap is controlled by local transition factors. From the exact mass partition and terminal identity $\Omega_q = 2\omega_{\text{end}} - 1$, we keep the full ordered reciprocal product unrelaxed and apply a two-state entropy inequality. Consecutive endpoint terms cancel, leaving only the two ends of the chain.

Contribution. The direct entropy estimate for the full ordered inverse product proves

$$d_q \leq \frac{1}{2} \quad \implies \quad \mathcal{H}_T(\alpha) \geq \frac{1}{5M_q}.$$

The neighboring-cutoff recurrences and a Lyapunov estimate at $\beta_\star = \sqrt{3}$ propagate this bound through large cutoffs; a finite-cutoff lemma and complete scan handle the rest. No optimality is claimed for the threshold, constant, or exponent.

Proof roadmap. Figure 1 records every numbered result and the two final transfer steps. Three scalar branches meet in Proposition 6.3; the geometric branch joins afterward to realize and rescale the hard instance.

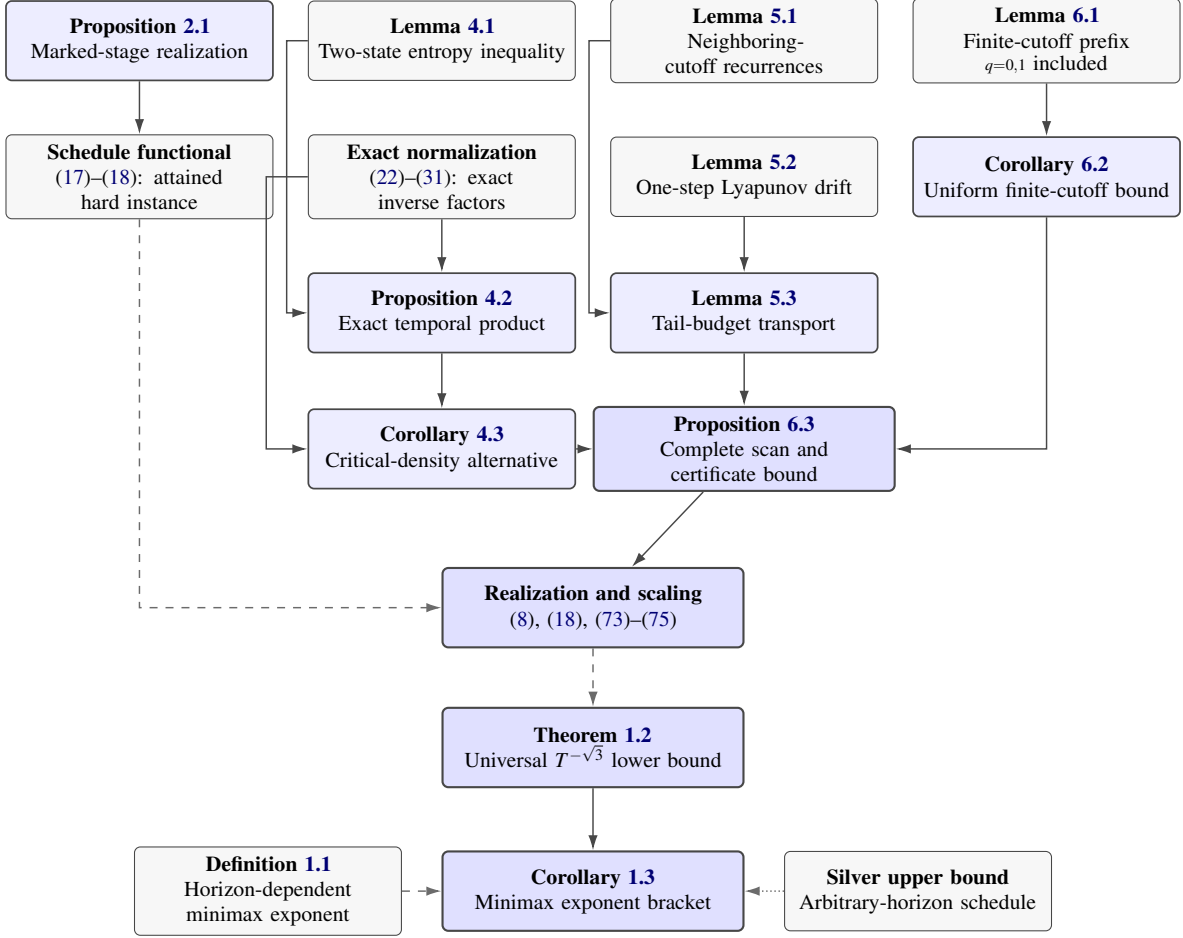


Figure 1: Proof architecture for Theorem 1.2 and Corollary 1.3. Solid arrows denote direct internal proof dependencies; dashed arrows denote realization, definition, or horizon/scaling transfers; and the dotted arrow denotes the external silver-schedule input. The full-product entropy branch is the product-control replacement introduced here; the realization, cutoff-dynamics, finite-cutoff, and scan branches have counterparts in Ma–Chen [13], but every solid or dashed dependency shown here is established inside this paper.

The realization branch, neighboring-cutoff recurrences, Lyapunov transport, and finite-cutoff branch have counterparts in the Ma–Chen framework [13]; their proofs below are complete and use no external Ma–Chen result. The full-product entropy branch is the replacement introduced here: it gives the half-density input to that outer framework. On its complement, the neighboring-cutoff recurrences and the one-step Lyapunov lemma jointly yield tail-budget transport. The finite-cutoff branch supplies the remaining finite cases. Corollary 4.3, Lemma 5.3, and Corollary 6.2 are exactly the three inputs scanned in Proposition 6.3.

The edge cases are localized in the diagram as follows. The empty marked chain $J = 0$ is included in (17); the exact-product branch is used only for $q \geq 2$; and Lemma 6.1 handles $q = 1$ with an empty internal product and $q = 0$ through the empty chain. The $N < q_*$ and small-horizon cases close through Corollary 6.2. In Lemma 5.3, $k = N$ is immediate and $k < N$ uses the core estimate at $k_0 = k + 1$. Thus the final scan requires neither an undefined empty product nor a limiting argument from a larger exponent.

2 The dimensionless certificate functional

We first set $L = R = 1$ and write the normalized schedule as

$$\alpha_t := L\eta_t, \quad 0 \leq t < T.$$

The construction below associates a hard convex 1-smooth function with any numerical vector $\alpha \in \mathbb{R}_{\geq 0}^T$. The capped-surplus stage data, realization, and maximized certificate in this section are notational counterparts of the construction in Sections 4.1–4.4 of Ma and Chen [13]. All claims needed from this construction are proved here, including the exact terminal scale and the empty-chain convention used later.

2.1 Step surpluses and stage data

Separate the capped and excess portions of each step:

$$\delta_t := (\alpha_t - 1)_+, \quad M_{\text{cap}} := 1 + \sum_{t=0}^{T-1} \min\{\alpha_t, 1\}, \quad N := |\{t : \delta_t > 0\}|. \quad (7)$$

In particular,

$$M_{\text{cap}} \leq T + 1, \quad N + 1 \leq T + 1. \quad (8)$$

Choose J surplus-bearing indices

$$0 \leq \iota_1 < \dots < \iota_J < T, \quad \delta_{\iota_j} > 0.$$

When $J \geq 1$, they split the unselected indices into gaps

$$\begin{aligned} \mathcal{J}_0 &= \{0, \dots, \iota_1 - 1\}, \\ \mathcal{J}_i &= \{\iota_i + 1, \dots, \iota_{i+1} - 1\}, \quad 1 \leq i < J, \\ \mathcal{J}_J &= \{\iota_J + 1, \dots, T - 1\}. \end{aligned}$$

For $J = 0$, set $\mathcal{J}_0 = \{0, \dots, T - 1\}$; all nonterminal quantities below are then absent. For $0 \leq i < J$, define

$$\omega_i := 1 + \sum_{t \in \mathcal{J}_i} \alpha_t, \quad \Omega_i := \omega_i + \delta_{\iota_{i+1}}, \quad (9)$$

and define the terminal scale

$$\Omega_J := 1 + 2 \sum_{t \in \mathcal{J}_J} \alpha_t. \quad (10)$$

The local amplitude bound at the transition from block i to block $i + 1$ is

$$\rho_i := \frac{\delta_{\iota_{i+1}} \Omega_{i+1}}{\omega_i (\Omega_i + \Omega_{i+1})}, \quad 0 \leq i < J. \quad (11)$$

For every J , the terminal scale Ω_J is positive. When $J \geq 1$, the quantities $\omega_i, \Omega_i, \rho_i$ are positive for $0 \leq i < J$. When $J = 0$, the only stage scale is $\Omega_0 = 1 + 2 \sum_{t=0}^{T-1} \alpha_t > 0$, and every product indexed by $0 \leq i < J$ is the empty product.

2.2 Geometric realization

The following proposition is the marked-stage Moreau-envelope realization corresponding to Ma and Chen's Theorem 4.1 and Section 4.3 [13]. We prove it below rather than invoke their result; the proposition explains why it is enough to study the scalar functional introduced next.

Proposition 2.1 (Marked-stage realization). *Fix $\alpha \in \mathbb{R}_{\geq 0}^T$, a marked chain $0 \leq \iota_1 < \dots < \iota_J < T$ as above, and positive numbers κ_i satisfying $0 < \kappa_i \leq \sqrt{\rho_i}$ for $0 \leq i < J$. Then there exist a convex 1-smooth $\Phi : \mathbb{R}^{J+1} \rightarrow \mathbb{R}$, a minimizer at the origin, and an initial point $\hat{x}_0$ with $\|\hat{x}_0\| = 1$, such that GD with schedule α satisfies*

$$\Phi(\hat{x}_T) - \Phi(0) = \frac{1}{2\Omega_J} \prod_{i=0}^{J-1} \kappa_i^2. \quad (12)$$

Proof. Let $\mathbf{u}_0, \dots, \mathbf{u}_J$ be orthonormal. Set

$$\sigma_0 = 1, \quad \sigma_{i+1} = \kappa_i \sigma_i, \quad P_i = \sigma_i \mathbf{u}_i,$$

and

$$p_i = \frac{\sigma_i}{\Omega_i} (\mathbf{u}_i - \kappa_i \mathbf{u}_{i+1}) \quad (0 \leq i < J), \quad p_J = \frac{\sigma_J}{\Omega_J} \mathbf{u}_J.$$

Let $\mathcal{P} = \text{conv}\{0, p_0, \dots, p_J\}$, let $s_{\mathcal{P}}(w) = \max_{p \in \mathcal{P}} \langle p, w \rangle$, and take the Moreau envelope

$$\Phi(x) := \min_w \left\{ s_{\mathcal{P}}(w) + \frac{1}{2} \|x - w\|^2 \right\}. \quad (13)$$

We first verify directly the convex-analysis facts needed below. For a nonempty closed convex set C , the Euclidean projection exists and is unique in finite dimensions (existence is immediate for the compact set $\mathcal{P}$, and uniqueness follows from strict convexity of the squared distance). It satisfies

$$\pi = \text{proj}_C(\xi) \iff \pi \in C \quad \text{and} \quad \langle \xi - \pi, p - \pi \rangle \leq 0 \quad \text{for every } p \in C. \quad (14)$$

Indeed, necessity follows by differentiating $\|\xi - \pi - s(p - \pi)\|^2$ at $s = 0^+$, since $\pi + s(p - \pi) \in C$ for $0 \leq s \leq 1$. Conversely, expanding $\|\xi - p\|^2$ and using the displayed inequality shows that it is at least $\|\xi - \pi\|^2$. Applying the criterion twice also gives

$$\|\text{proj}_C(\xi) - \text{proj}_C(\xi')\|^2 \leq \langle \text{proj}_C(\xi) - \text{proj}_C(\xi'), \xi - \xi' \rangle, \quad (15)$$

so the projection is nonexpansive.

Now let $\pi = \text{proj}_{\mathcal{P}}(x)$ and $w_0 = x - \pi$. Criterion (14) gives $s_{\mathcal{P}}(w_0) = \langle \pi, w_0 \rangle$. For arbitrary w ,

$$\begin{aligned} s_{\mathcal{P}}(w) + \frac{1}{2} \|x - w\|^2 &\geq \langle \pi, w \rangle + \frac{1}{2} \|x - w\|^2 \\ &= \langle \pi, x \rangle - \frac{1}{2} \|\pi\|^2 + \frac{1}{2} \|w - (x - \pi)\|^2. \end{aligned}$$

Equality holds at $w = w_0$, and hence the identity $\langle p, x \rangle - \|p\|^2/2 = \|x\|^2/2 - \|x - p\|^2/2$ also shows that

$$\Phi(x) = \langle \pi, x \rangle - \frac{1}{2} \|\pi\|^2 = \frac{1}{2} \|x\|^2 - \frac{1}{2} \text{dist}(x, \mathcal{P})^2 = \max_{p \in \mathcal{P}} \left\{ \langle p, x \rangle - \frac{1}{2} \|p\|^2 \right\}. \quad (16)$$

The maximum representation proves convexity. If $\pi_x = \text{proj}_{\mathcal{P}}(x)$ and $\pi_y = \text{proj}_{\mathcal{P}}(y)$, it also gives

$$0 \leq \Phi(y) - \Phi(x) - \langle \pi_x, y - x \rangle \leq \langle \pi_y - \pi_x, y - x \rangle \leq \|y - x\|^2.$$

The displayed remainder is $O(\|y - x\|^2)$, so Φ is differentiable at x with $\nabla\Phi(x) = \pi_x$. Moreover, (15) shows that this gradient is 1-Lipschitz. Finally, $0 \in \mathcal{P}$ makes the maximum in (16) nonnegative for every x and equal to zero at $x = 0$. Therefore Φ is convex and 1-smooth, and 0 is a minimizer.

Because $\mathcal{P}$ is the convex hull of $0, p_0, \dots, p_J$, it is enough to check the displayed inequality at those vertices. Equivalently, with $v = \xi - \pi$, the candidate π must maximize $\langle p, v \rangle$ over the listed vertices.

We now verify the intended projections. On a nonterminal block, write

$$Y_i(\tau) := P_i - \tau p_i, \quad 0 \leq \tau \leq \omega_i - 1,$$

put $a = \tau + 1$, and observe that

$$W_{i,a} := Y_i(\tau) - p_i = \frac{\sigma_i}{\Omega_i} ((\Omega_i - a)\mathbf{u}_i + a\kappa_i\mathbf{u}_{i+1}).$$

The vector $W_{i,a}$ is supported on $\text{span}\{\mathbf{u}_i, \mathbf{u}_{i+1}\}$. For $j < J$, the vector p_j is supported on $\text{span}\{\mathbf{u}_j, \mathbf{u}_{j+1}\}$, while the terminal vector p_J is supported on $\text{span}\{\mathbf{u}_J\}$. Thus, among the existing vertices, only p_{i-1}, p_i, p_{i+1} can have a nonzero inner product with $W_{i,a}$, and

$$\begin{aligned} \langle p_{i-1}, W_{i,a} \rangle &= -\frac{\sigma_i^2(\Omega_i - a)}{\Omega_{i-1}\Omega_i} \leq 0 \quad (i > 0), \\ \langle p_i, W_{i,a} \rangle &= \frac{\sigma_i^2}{\Omega_i^2} \{\Omega_i - a(1 + \kappa_i^2)\}, \\ \langle p_{i+1}, W_{i,a} \rangle &= \frac{\sigma_i^2 \kappa_i^2 a}{\Omega_i \Omega_{i+1}}. \end{aligned}$$

Here $\Omega_i - a \geq \Omega_i - \omega_i = \delta_{i+1} > 0$, which justifies the displayed sign of the p_{i-1} term. The middle expression decreases in a , while the last increases. At $a = \omega_i$, their comparison is exactly

$$\langle p_i, W_{i,\omega_i} \rangle \geq \langle p_{i+1}, W_{i,\omega_i} \rangle \iff \kappa_i^2 \leq \frac{\delta_{i+1} \Omega_{i+1}}{\omega_i(\Omega_i + \Omega_{i+1})} = \rho_i.$$

The same comparison therefore holds for every $1 \leq a \leq \omega_i$. Moreover, $\langle p_{i+1}, W_{i,a} \rangle \geq 0$, so the candidate p_i also dominates the vertex 0 and every nonneighboring vertex, whose inner products vanish; it dominates p_{i-1} because the latter inner product is nonpositive. Criterion (14) now gives $\text{proj}_{\mathcal{P}}(Y_i(\tau)) = p_i$.

On the terminal block, put

$$\tau_{\text{end}} := \sum_{t \in \mathcal{J}_J} \alpha_t, \quad Y_J(\tau) := P_J - \tau p_J, \quad 0 \leq \tau \leq \tau_{\text{end}}.$$

Then

$$Y_J(\tau) - p_J = \frac{\sigma_J(\Omega_J - \tau - 1)}{\Omega_J} \mathbf{u}_J$$

is a nonnegative multiple of $\mathbf{u}_J$, since $\Omega_J - \tau - 1 = 2\tau_{\text{end}} - \tau \geq 0$. Its inner product with p_J is nonnegative, that with p_{J-1} is nonpositive when $J > 0$, and all other vertex inner products vanish. In particular, p_J also dominates the vertex 0. Criterion (14) gives $\text{proj}_{\mathcal{P}}(Y_J(\tau)) = p_J$.

Starting from $\hat{x}_0 = P_0 = \mathbf{u}_0$, for each $0 \leq i < J$ consider first the unselected gap steps in $\mathcal{J}_i$. At each such query, τ is the cumulative stepsize of the preceding gap steps and lies in $[0, \omega_i - 1]$, so the gradient is the constant vector p_i . After all gap steps the iterate is $Y_i(\omega_i - 1)$. The selected long step then has size $1 + \delta_{i+1}$, uses the same gradient p_i , and produces

$$P_i - \Omega_i p_i = \sigma_i \kappa_i \mathbf{u}_{i+1} = P_{i+1}.$$

After the last selected transition, or immediately when $J = 0$, the terminal gap steps follow $Y_J(\tau)$ and keep gradient p_J for $0 \leq \tau \leq \tau_{\text{end}}$. This proves the claimed trajectory through all blocks, including the terminal block. At the final point, $\text{proj}_{\mathcal{D}}(\hat{x}_T) = p_J$, and direct substitution in (13) gives

$$\begin{aligned}\Phi(\hat{x}_T) - \Phi(0) &= \langle \hat{x}_T, p_J \rangle - \frac{1}{2} \|p_J\|^2 \\ &= \frac{\sigma_J^2}{\Omega_J^2} \left(\Omega_J - \tau_{\text{end}} - \frac{1}{2} \right) = \frac{\sigma_J^2}{2\Omega_J} = \frac{1}{2\Omega_J} \prod_{i=0}^{J-1} \kappa_i^2.\end{aligned}$$

The penultimate equality uses $\Omega_J = 1 + 2\tau_{\text{end}}$. □

The same maximized functional appears in the Ma–Chen reduction [13, Equation (20) and Corollary 4.3]. Here its validity follows directly from Proposition 2.1: maximizing over all marked chains and choosing $\kappa_i^2 = \rho_i$ defines

$$\mathcal{H}_T(\alpha) := \max_{0 \leq J \leq N} \max_{\substack{0 \leq t_1 < \dots < t_J < T \\ \delta_{t_j} > 0}} \frac{1}{\Omega_J} \prod_{i=0}^{J-1} \rho_i. \quad (17)$$

When $J = 0$, the inner maximum consists of the unique empty chain and the product equals 1. Proposition 2.1 yields an instance of dimension at most $T + 1$ satisfying

$$2\{\Phi(\hat{x}_T) - \Phi(0)\} = \mathcal{H}_T(\alpha). \quad (18)$$

The rest of the proof is therefore purely a lower bound on $\mathcal{H}_T(\alpha)$.

3 Ordered surpluses and the exact temporal product

3.1 Cutoff statistics

The ranked-surplus and residual-budget variables below correspond to those in Sections 5.1, 5.4, and 6.1 of Ma and Chen [13]. In particular, M_q, d_q, g_q correspond to their D_q, ζ_q, v_q , respectively. All properties used here are derived from the definitions below. Let $t_{[1]}, \dots, t_{[N]}$ be the positive-surplus time indices, ordered so that

$$\delta_{t_{[1]}} \geq \delta_{t_{[2]}} \geq \dots \geq \delta_{t_{[N]}} > 0.$$

Break ties once and for all, say by increasing time, and abbreviate $\delta_{[j]} := \delta_{t_{[j]}}$. When $N = 0$, this ranked list is empty. For $0 \leq q \leq N$, define the residual budget

$$M_q := M_{\text{cap}} + \sum_{j=q+1}^N \delta_{[j]}. \quad (19)$$

For $1 \leq q \leq N$, define

$$d_q := \frac{M_q}{q^2} \sum_{j=1}^q \frac{1}{\delta_{[j]}}, \quad g_q := \frac{q\delta_{[q]}}{M_q}. \quad (20)$$

The elementary density relation

$$d_q g_q = \frac{\delta_{[q]}}{q} \sum_{j=1}^q \frac{1}{\delta_{[j]}} \leq 1 \quad (21)$$

will later convert a lower bound on d_q into an upper bound on g_q .

3.2 Temporal normalization

The same chronological restoration and mass partition appear in Ma–Chen [13, Section 5.1 and Equation (24)]. We derive them directly below from the schedule. The exact terminal factor is shared as well. Our departure begins with the normalized states in (23)–(24): we retain the full ordered product and control it directly, rather than passing from the exact local identities to budget equalization and matching bounds.

Fix $2 \leq q \leq N$. Let $t_1^{(q)} < \dots < t_q^{(q)}$ be the increasing rearrangement of $t_{[1]}, \dots, t_{[q]}$, and set

$$\widehat{\delta}_i := \delta_{t_i^{(q)}}, \quad 1 \leq i \leq q.$$

Let $\omega_0, \dots, \omega_{q-1}$ be the nonterminal gap masses from (9) for this marked chain, and define

$$\omega_{\text{end}} := 1 + \sum_{t \text{ after the last selected index}} \alpha_t.$$

Each selected long step contributes its capped unit to M_{cap} but not its excess to M_q ; each unselected step contributes its full value after the residual excesses are added to M_{cap} . Hence

$$M_q = 1 + q + \sum_{t \text{ not selected}} \alpha_t.$$

The q constants in $\omega_0, \dots, \omega_{q-1}$ account for the selected capped units, and the constant in ω_{end} accounts for the leading 1. This gives the exact mass and terminal identities

$$\sum_{i=0}^{q-1} \omega_i + \omega_{\text{end}} = M_q, \quad \Omega_q = 2\omega_{\text{end}} - 1. \quad (22)$$

Write $\bar{m} := M_q/q$, and introduce the dimensionless states

$$b_i := \frac{\omega_{i-1}}{\bar{m}}, \quad u_i := \frac{2\bar{m}}{\widehat{\delta}_i}, \quad r_i := b_i + \frac{2}{u_i}, \quad 1 \leq i \leq q. \quad (23)$$

Also put

$$b_\infty := \frac{\omega_{\text{end}}}{\bar{m}} = q - \sum_{i=1}^q b_i, \quad \varepsilon_\infty := \frac{1}{\bar{m}} = \frac{q}{M_q}, \quad r_\infty := 2b_\infty - \varepsilon_\infty. \quad (24)$$

All these variables are positive: $\omega_i, \omega_{\text{end}}, \bar{m}, \widehat{\delta}_i > 0$, so $b_i, u_i, r_i, b_\infty, \varepsilon_\infty > 0$. Moreover, $\omega_{\text{end}} \geq 1$ gives $b_\infty = \omega_{\text{end}}/\bar{m} \geq 1/\bar{m} = \varepsilon_\infty$, while $b_\infty = q - \sum_i b_i < q$. Therefore

$$0 < r_\infty < 2b_\infty < 2q. \quad (25)$$

The definitions give the two exact identities

$$\sum_{i=1}^q u_i = \frac{2M_q}{q} \sum_{i=1}^q \frac{1}{\widehat{\delta}_i} = 2qd_q, \quad (26)$$

and

$$\Omega_{i-1} = \bar{m}r_i \quad (1 \leq i \leq q), \quad \Omega_q = \bar{m}r_\infty. \quad (27)$$

3.3 Exact reciprocal transitions

The local and terminal reciprocal identities have counterparts in Equation (26) of Ma–Chen [13]; equations (28)–(29) derive them directly from the stage definitions. The global state normalization and inverse product in (30) then give a different product estimate.

For $1 \leq i < q$, equation (11) reads

$$\rho_{i-1} = \frac{\widehat{\delta}_i \Omega_i}{\omega_{i-1}(\Omega_{i-1} + \Omega_i)}.$$

Using (23)–(27), its reciprocal is

$$R_i := \rho_{i-1}^{-1} = \frac{b_i u_i}{2} \frac{r_i + r_{i+1}}{r_{i+1}}. \quad (28)$$

At the last edge the exact terminal scale $\Omega_q = 2\omega_{\text{end}} - 1$ must be retained:

$$\begin{aligned} \Omega_q \rho_{q-1}^{-1} &= \frac{\omega_{q-1}(\Omega_{q-1} + \Omega_q)}{\widehat{\delta}_q} \\ &= \bar{m} \frac{b_q u_q}{2} (r_q + r_\infty). \end{aligned} \quad (29)$$

In particular, the finite -1 in $\Omega_q = 2\omega_{\text{end}} - 1$ survives as the $-\varepsilon_\infty$ correction in $r_\infty = 2b_\infty - \varepsilon_\infty$. No asymptotic replacement of the terminal scale is made.

Define the dimensionless inverse-chain product

$$\Pi_q := \frac{1}{2M_q} \left(\Omega_q \prod_{j=0}^{q-1} \rho_j^{-1} \right) = \frac{b_q u_q (r_q + r_\infty)}{4q} \prod_{i=1}^{q-1} R_i. \quad (30)$$

Every factor in (30) is positive. Since the marked chain is one of the chains maximized over in (17), inversion of (30) gives

$$\mathcal{H}_T(\alpha) \geq \frac{1}{\Omega_q} \prod_{j=0}^{q-1} \rho_j = \frac{1}{2M_q \Pi_q}. \quad (31)$$

Equations (28)–(31) contain no budget equalization, auxiliary terminal vertex, or matching relaxation of the type used in Ma–Chen [13]; nor do they replace either adjacent block scale by a lower bound.

4 Entropy control and telescoping

The local identities used here were proved in Section 3. Ma–Chen passes from the corresponding identities to a two-matching relaxation [13, Secs. 5.2–5.4]; this section instead proves a direct full-path entropy bound for the ordered product.

Let

$$\text{ent}(t) := t - 1 - \log t, \quad t > 0. \quad (32)$$

This nonnegative entropy measures the deviation of t from 1.

Lemma 4.1 (Two-state block-scale inequality). *For arbitrary $b, u, \tilde{b}, \tilde{u} > 0$, let*

$$r = b + \frac{2}{u}, \quad \tilde{r} = \tilde{b} + \frac{2}{\tilde{u}}, \quad e = \text{ent}(b) + \text{ent}(u), \quad \tilde{e} = \text{ent}(\tilde{b}) + \text{ent}(\tilde{u}).$$

Then

$$\frac{r + \tilde{r}}{2\sqrt{r\tilde{r}}} \leq \exp\left(\frac{e + \tilde{e}}{2}\right). \quad (33)$$

Proof. For $a, c > 0$, setting $\bar{a} = (a + c)/2$ gives

$$e^{-\{\text{ent}(a) + \text{ent}(c)\}/2} \frac{a + c}{2\sqrt{ac}} = \bar{a}e^{1-\bar{a}} \leq 1. \quad (34)$$

Put

$$c_b = e^{-\{\text{ent}(b) + \text{ent}(\tilde{b})\}/2}, \quad c_u = e^{-\{\text{ent}(u) + \text{ent}(\tilde{u})\}/2}.$$

Both are at most one. Applying (34) to $b, \tilde{b}$ and to $u, \tilde{u}$ gives

$$c_b(b + \tilde{b}) \leq 2\sqrt{b\tilde{b}}, \quad c_u(u^{-1} + \tilde{u}^{-1}) \leq \frac{2}{\sqrt{u\tilde{u}}}.$$

Therefore

$$\begin{aligned} c_b c_u(r + \tilde{r}) &= c_b c_u \{b + \tilde{b} + 2(u^{-1} + \tilde{u}^{-1})\} \\ &\leq 2c_u \sqrt{b\tilde{b}} + \frac{4c_b}{\sqrt{u\tilde{u}}} \\ &\leq 2\sqrt{b\tilde{b}} + \frac{4}{\sqrt{u\tilde{u}}} \\ &\leq 2\sqrt{(b + 2/u)(\tilde{b} + 2/\tilde{u})}. \end{aligned}$$

The final inequality follows after squaring from

$$\frac{2b}{\tilde{u}} + \frac{2\tilde{b}}{u} \geq 4\sqrt{\frac{b\tilde{b}}{u\tilde{u}}}.$$

Since $c_b c_u = e^{-(e+\tilde{e})/2}$, this is (33). □

For the state (b_i, u_i) , write

$$e_i := \text{ent}(b_i) + \text{ent}(u_i). \quad (35)$$

Combining Lemma 4.1 with (28) gives

$$R_i \leq b_i u_i \exp\left(\frac{e_i + e_{i+1}}{2}\right) \sqrt{\frac{r_i}{r_{i+1}}}. \quad (36)$$

The identity

$$\log(b_i u_i) = b_i + u_i - 2 - e_i \quad (37)$$

then makes the path product telescope:

$$\prod_{i=1}^{q-1} R_i \leq \exp\left\{\sum_{i=1}^{q-1} (b_i + u_i - 2) + \frac{e_q - e_1}{2}\right\} \sqrt{\frac{r_1}{r_q}}. \quad (38)$$

The entropy exponent cancels exactly because

$$-\sum_{i=1}^{q-1} e_i + \frac{1}{2} \sum_{i=1}^{q-1} (e_i + e_{i+1}) = \frac{e_q - e_1}{2}.$$

Thus every interior term $e_2, \dots, e_{q-1}$ disappears and only the two boundary states remain.

Define

$$E_i^+ := r_i e^{-e_i}, \quad E_i^- := r_i^{-1} e^{-e_i}. \quad (39)$$

Substituting (38) into (30) and applying (37) once more at $i = q$ yields

$$\Pi_q \leq \exp \left\{ \sum_{i=1}^q (b_i + u_i - 2) \right\} \frac{\sqrt{E_1^+ E_q^+} + r_\infty \sqrt{E_1^+ E_q^-}}{4q}. \quad (40)$$

The endpoint factor is uniformly bounded. The following estimates are deliberately loose, and no optimality of their numerical constants is claimed. For arbitrary $b, u > 0$,

$$\begin{aligned} E^+ &= \left(b + \frac{2}{u} \right) e^{-\text{ent}(b) - \text{ent}(u)} = b(bu + 2)e^{2-b-u} \\ &\leq \frac{4}{e} + 2e < 8. \end{aligned} \quad (41)$$

Here we used

$$\sup_{t>0} t^2 e^{-t} = \frac{4}{e^2}, \quad \sup_{t>0} t e^{-t} = \frac{1}{e}, \quad \sup_{t>0} e^{-t} = 1.$$

Similarly, using $bu + 2 \geq 2$,

$$E^- = \frac{bu^2 e^{2-b-u}}{bu + 2} \leq \frac{2}{e} < 1. \quad (42)$$

Since $q \geq 2$ and $r_\infty < 2q$, equations (40)–(42) imply

$$\frac{\sqrt{E_1^+ E_q^+} + r_\infty \sqrt{E_1^+ E_q^-}}{4q} < \frac{8}{4q} + \frac{2q\sqrt{8}}{4q} \leq 1 + \sqrt{2} < \frac{5}{2}. \quad (43)$$

We have proved the following global product bound.

Proposition 4.2 (Exact temporal product). *Fix $\alpha \in \mathbb{R}_{\geq 0}^T$, and define the cutoff statistics and normalized states by (19)–(24). For every $2 \leq q \leq N$,*

$$\Pi_q \leq \frac{5}{2} \exp \left\{ \sum_{i=1}^q (b_i + u_i - 2) \right\}. \quad (44)$$

Corollary 4.3 (Half-density certificate). *Under the hypotheses and notation of Proposition 4.2, every $2 \leq q \leq N$ satisfies*

$$d_q \leq \frac{1}{2} \implies \mathcal{H}_T(\alpha) \geq \frac{1}{5M_q}. \quad (45)$$

Proof. Equations (24) and (26) give

$$\sum_{i=1}^q b_i = q - b_\infty, \quad \sum_{i=1}^q u_i = 2qd_q \leq q.$$

Consequently,

$$\sum_{i=1}^q (b_i + u_i - 2) \leq -b_\infty < 0.$$

Proposition 4.2 gives $\Pi_q \leq 5/2$, and (31) finishes the proof. $\square$

Remark 4.4 (Why a one-edge Jensen argument is insufficient). The exact four-variable edge factor is

$$\mathcal{Q}_{\text{edge}}(u, \tilde{u}, b, \tilde{b}) = \frac{b\{2(u + \tilde{u}) + u\tilde{u}(b + \tilde{b})\}}{2(2 + \tilde{b}\tilde{u})}.$$

Even on the equal-gap slice $b = \tilde{b}$, the logarithm of

$$\mathcal{Q}_{\text{diag}}(u, \tilde{u}, b) = \frac{b(u + \tilde{u} + bu\tilde{u})}{2 + b\tilde{u}}$$

is not jointly concave. At $(u, \tilde{u}, b) = (1/10, 10, 1)$, for example,

$$(1, 0, 1) \nabla^2 \log \mathcal{Q}_{\text{diag}}|_{(1/10, 10, 1)} (1, 0, 1)^T = \frac{16141}{49284} > 0.$$

This failure of local concavity motivates the full-path estimate: entropy paid at one endpoint of an edge is recovered at the next edge, producing the cancellation in (38).

5 Cutoff dynamics and endpoint Lyapunov estimate

We next establish the cutoff dynamics needed to use the half-density certificate from Cor. 4.3. These dynamics have counterparts in Ma–Chen [13], but every identity and estimate needed here is proved below. Set

$$\beta_\star := \sqrt{3}, \quad d_\star := \frac{1}{2}, \quad q_\star := 3. \quad (46)$$

The critical identity is

$$d_\star^{-1} = 2 = (\beta_\star - 1)(\beta_\star + 1) < q_\star. \quad (47)$$

The parameters are fixed at these endpoint values throughout; no argument passes to a limit from an exponent larger than $\sqrt{3}$.

5.1 Exact recurrences

The same neighboring-cutoff identities appear in Ma–Chen [13, Lemma 6.2 and Appendix B.2]. The proof immediately below derives them from (19)–(20), with all index ranges explicit.

Lemma 5.1 (Neighboring cutoffs). *Fix $\alpha \in \mathbb{R}_{\geq 0}^T$ and define M_q, d_q, g_q by (19)–(20). For $1 \leq q \leq N$,*

$$\frac{M_{q-1}}{M_q} = 1 + \frac{g_q}{q}. \quad (48)$$

Moreover, for every $0 \leq k \leq N$,

$$\frac{M_k}{M_{\text{cap}}} = \prod_{j=k+1}^N \left(1 + \frac{g_j}{j}\right). \quad (49)$$

The product is empty when $k = N$. For $1 \leq q < N$,

$$\frac{g_q}{q} \left(1 + \frac{g_{q+1}}{q+1}\right) \geq \frac{g_{q+1}}{q+1}, \quad (50)$$

and, whenever $q > g_q$,

$$g_{q+1} \leq \frac{(q+1)g_q}{q - g_q}. \quad (51)$$

Finally,

$$d_{q+1} = \frac{q^2 d_q}{(q+1)(q+1+g_{q+1})} + \frac{1}{(q+1)g_{q+1}}. \quad (52)$$

Proof. The identity $M_{q-1} = M_q + \delta_{[q]}$ and $\delta_{[q]}/M_q = g_q/q$ prove the first relation. Multiplication and $M_N = M_{\text{cap}}$ prove the product formula. Next, $\delta_{[q]} \geq \delta_{[q+1]}$ gives

$$\frac{g_q}{q} = \frac{\delta_{[q]}}{M_q} \geq \frac{\delta_{[q+1]}}{M_q} = \frac{g_{q+1}/(q+1)}{1 + g_{q+1}/(q+1)},$$

which is (50); rearrangement gives $g_{q+1}(q - g_q) \leq (q+1)g_q$ and hence (51).

Finally,

$$\sum_{j=1}^{q+1} \frac{1}{\delta_{[j]}} = \frac{q^2 d_q}{M_q} + \frac{q+1}{g_{q+1} M_{q+1}}.$$

Multiplication by $M_{q+1}/(q+1)^2$, followed by

$$\frac{M_{q+1}}{M_q} = \left(1 + \frac{g_{q+1}}{q+1}\right)^{-1},$$

proves (52). □

5.2 The one-step potential

The Lyapunov weight and coefficient-comparison template have counterparts in Ma–Chen [13, Lemma 6.3 and Appendix B.3]. Lemma 5.2 and its proof rederive the needed inequality at the endpoint parameters in (46), which lie outside the exponent range of their stated theorem.

Define

$$w(z) := \frac{z}{d_*(z + \beta_* + 1)}, \quad V_q := w(g_q)(d_q - d_*), \quad (53)$$

and choose

$$\Gamma_{\text{dr}} := \max \left\{ 1, \frac{d_*^{-1} \{1 + d_*^{-1}(d_*^{-1} + 2)\}}{\beta_* + 1} \right\}. \quad (54)$$

At (46),

$$\Gamma_{\text{dr}} = \frac{18}{1 + \sqrt{3}}.$$

Lemma 5.2 (One-step Lyapunov drift). *Fix $\beta_* = \sqrt{3}$ and $d_* = 1/2$ as in (46). Let $n - 1 > d_*^{-1}$, and suppose*

$$d_{n-1} \geq d_*, \quad 0 < g_{n-1}, g_n \leq d_*^{-1},$$

together with the transition and recursion

$$g_n \leq \frac{ng_{n-1}}{n-1-g_{n-1}}, \quad d_n = \frac{(n-1)^2 d_{n-1}}{n(n+g_n)} + \frac{1}{ng_n}.$$

Then

$$V_n - V_{n-1} \leq \frac{\beta_* - 1 - g_n}{n} + \frac{\Gamma_{\text{dr}}}{n^2}. \quad (55)$$

Proof. Write $g_- := g_{n-1}$, $g_+ := g_n$, and

$$t_n(g_+) := \frac{(n-1)^2}{n(n+g_+)}.$$

The hypotheses give $n - 1 - g_{n-1} > 0$, so the denominator in the assumed transition is positive. Solving that transition for g_- gives

$$g_- \geq g_{\text{low}} := \frac{(n-1)g_+}{n+g_+}.$$

Moreover,

$$w'(z) = \frac{\beta_* + 1}{d_*(z + \beta_* + 1)^2} > 0, \quad \frac{w(z)}{z} = \frac{1}{d_*(z + \beta_* + 1)}$$

is decreasing. Since $0 < g_{\text{low}} < g_+$,

$$\begin{aligned} w(g_-) &\geq w(g_{\text{low}}) \geq \frac{g_{\text{low}}}{g_+} w(g_+) \\ &= \frac{n-1}{n+g_+} w(g_+) \geq t_n(g_+) w(g_+). \end{aligned} \tag{56}$$

Subtracting d_* from the d -recursion gives

$$d_n - d_* = t_n(g_+)(d_{n-1} - d_*) + \varepsilon_n(g_+), \tag{57}$$

where

$$\varepsilon_n(g_+) := \frac{1}{ng_+} - \frac{d_*\{n(g_+ + 2) - 1\}}{n(n+g_+)}.$$

Using $d_*^{-1} = (\beta_* - 1)(\beta_* + 1)$, direct simplification gives the exact identity

$$w(g_+)\varepsilon_n(g_+) = \frac{1}{n} \left[\beta_* - 1 - g_+ + \frac{g_+\{1 + g_+(g_+ + 2)\}}{(n+g_+)(g_+ + \beta_* + 1)} \right]. \tag{58}$$

For verification, subtract $\beta_* - 1 - g_+$ from $nw(g_+)\varepsilon_n(g_+)$; the remainder is

$$\frac{g_+}{g_+ + \beta_* + 1} \left\{ (g_+ + 2) - \frac{n(g_+ + 2) - 1}{n+g_+} \right\} = \frac{g_+\{1 + g_+(g_+ + 2)\}}{(n+g_+)(g_+ + \beta_* + 1)}.$$

Because $d_{n-1} - d_* \geq 0$, equations (56) and (57) imply

$$\begin{aligned} V_n &= w(g_+)t_n(g_+)(d_{n-1} - d_*) + w(g_+)\varepsilon_n(g_+) \\ &\leq V_{n-1} + w(g_+)\varepsilon_n(g_+). \end{aligned}$$

Finally, $0 < g_+ \leq d_*^{-1}$ yields

$$\begin{aligned} 0 &\leq \frac{g_+\{1 + g_+(g_+ + 2)\}}{(n+g_+)(g_+ + \beta_* + 1)} \\ &= \frac{1}{n} \frac{n}{n+g_+} \frac{g_+}{g_+ + \beta_* + 1} \{1 + g_+(g_+ + 2)\} \\ &\leq \frac{\Gamma_{\text{dr}}}{n}. \end{aligned}$$

Combining this with (58) proves (55). □

5.3 Transport across a tail of cutoffs

The organization of the tail argument parallels the Ma–Chen Lyapunov telescope [13, Lemma 6.4]. The proof below starts only from the stated hypotheses and the internally proved Lemmas 5.1 and 5.2.

Define the finite universal constant

$$\Gamma_{\text{prop}} := 2 \exp \left\{ \frac{1}{d_*(\beta_* + 1)} + \frac{\pi^2}{6} \Gamma_{\text{dr}} \right\}. \quad (59)$$

Lemma 5.3 (Tail-budget transport). *Fix $\beta_* = \sqrt{3}$, $d_* = 1/2$, and $q_* = 3$ as in (46). Let $q_* \leq k \leq N$. If*

$$d_q > d_* \quad (k < q \leq N),$$

then

$$M_k k^{\beta_* - 1} \leq \Gamma_{\text{prop}} M_{\text{cap}} N^{\beta_* - 1}. \quad (60)$$

Proof. We first prove a core estimate. Suppose $q_* \leq k_0 \leq N$ and $d_q > d_*$ for every $k_0 \leq q \leq N$. By (21),

$$0 < g_q < d_*^{-1}.$$

For $n = k_0 + 1, \dots, N$, we have $n - 1 \geq q_* > d_*^{-1} > g_{n-1}$. Lemma 5.1 supplies the hypotheses of Lemma 5.2.

On this interval the potential is nonnegative, and

$$0 \leq V_q \leq \frac{g_q d_q}{d_*(g_q + \beta_* + 1)} \leq \frac{1}{d_*(\beta_* + 1)}. \quad (61)$$

Rearrange (55), sum from $n = k_0 + 1$ to N , and telescope:

$$\begin{aligned} \sum_{n=k_0+1}^N \frac{g_n}{n} &\leq (\beta_* - 1) \sum_{n=k_0+1}^N \frac{1}{n} + \Gamma_{\text{dr}} \sum_{n=k_0+1}^N \frac{1}{n^2} + V_{k_0} - V_N \\ &\leq (\beta_* - 1) \log \frac{N}{k_0} + \frac{1}{d_*(\beta_* + 1)} + \Gamma_{\text{dr}} \frac{\pi^2}{6}. \end{aligned} \quad (62)$$

Using (49) and $\log(1+t) \leq t$,

$$\log \frac{M_{k_0}}{M_{\text{cap}}} \leq \sum_{n=k_0+1}^N \frac{g_n}{n}.$$

Exponentiation of (62) gives

$$M_{k_0} k_0^{\beta_* - 1} \leq \frac{\Gamma_{\text{prop}}}{2} M_{\text{cap}} N^{\beta_* - 1}. \quad (63)$$

If $k = N$, then $M_k = M_{\text{cap}}$ and (60) is immediate. If $k < N$, apply (63) with $k_0 = k + 1$. Furthermore, $d_{k+1} > d_*$ implies $g_{k+1} < d_*^{-1}$, so

$$\frac{M_k}{M_{k+1}} = 1 + \frac{g_{k+1}}{k+1} < 1 + \frac{d_*^{-1}}{q_*} < 2.$$

Since $k^{\beta_* - 1} \leq (k+1)^{\beta_* - 1}$, the factor 2 converts (63) into (60). $\square$

6 Finite cutoffs and the complete scan

The bounded-rank prefix estimate below has a counterpart in Ma–Chen [13, Lemma B.1]. We prove it in full rather than invoke that lemma, with the labeling and empty-product conventions stated explicitly in our notation.

Lemma 6.1 (Finite-cutoff prefix certificate). *Fix $\alpha \in \mathbb{R}_{\geq 0}^T$, define M_q by (19), and fix $0 \leq q_a \leq N$. Among the chains formed by the global rank prefixes of time indices*

$$\emptyset, \quad \{t_{[1]}\}, \quad \{t_{[1]}, t_{[2]}\}, \quad \dots, \quad \{t_{[1]}, \dots, t_{[q_a]}\},$$

with every nonempty prefix restored to temporal order, at least one has certificate contribution $\Omega_J^{-1} \prod_{i=0}^{J-1} \rho_i$ in (17) at least

$$\frac{1}{4M_{q_a} q_a 8^{q_a-1}} \quad (q_a \geq 1). \quad (64)$$

For $q_a = 0$, the empty chain has contribution at least $(2M_0)^{-1}$.

Proof. The empty-chain value is

$$\frac{1}{1 + 2 \sum_t \alpha_t} = \frac{1}{2M_0 - 1} \geq \frac{1}{2M_0}.$$

Assume $q_a \geq 1$. Call a rank $j \in \{1, \dots, q_a\}$ *loaded* when $j\delta_{[j]} \geq M_j$, and *light* otherwise, equivalently when $j\delta_{[j]} < M_j$.

If every rank $1 \leq j \leq q_a$ is light, then

$$\frac{M_0}{M_{q_a}} = \prod_{j=1}^{q_a} \left(1 + \frac{\delta_{[j]}}{M_j}\right) < \prod_{j=1}^{q_a} \left(1 + \frac{1}{j}\right) = q_a + 1.$$

The empty chain is therefore at least

$$\frac{1}{2M_{q_a}(q_a + 1)} \geq \frac{1}{4M_{q_a} q_a 8^{q_a-1}}.$$

Otherwise, let q be the largest loaded rank. All ranks $q < j \leq q_a$ are light, so

$$M_q \leq M_{q_a} \frac{q_a + 1}{q + 1}, \quad \delta_{[q]} \geq \frac{M_q}{q}. \quad (65)$$

Select the time indices $t_{[1]}, \dots, t_{[q]}$, order them increasingly, and denote their surpluses by $\widehat{\delta}_1, \dots, \widehat{\delta}_q$. Let $\delta_{\min} = \widehat{\delta}_{[q]}$. The exact mass partition is

$$\sum_{i=0}^{q-1} \omega_i + \omega_{\text{end}} = M_q.$$

For $1 \leq i < q$, the unnormalized inverse identity is

$$\rho_{i-1}^{-1} = \frac{\omega_{i-1}}{\widehat{\delta}_i} + \frac{\omega_{i-1}}{\Omega_i} + \frac{\omega_{i-1}^2}{\widehat{\delta}_i \Omega_i}. \quad (66)$$

Since $\widehat{\delta}_i \geq \delta_{\min}$ and $\Omega_i = \omega_i + \widehat{\delta}_{i+1} \geq \delta_{\min}$ for $i < q$, this is bounded by

$$h(\omega_{i-1}/\delta_{\min}), \quad h(x) := x(x+2).$$

Moreover,

$$\sum_{i=0}^{q-2} \frac{\omega_i}{\delta_{\min}} \leq \frac{M_q}{\delta_{\min}} \leq q.$$

The function $\log h$ is concave because

$$\frac{d^2}{dx^2} \log h(x) = -\frac{1}{x^2} - \frac{1}{(x+2)^2} < 0.$$

If $q = 1$, the product below is empty and the claimed bound is immediate. For $q \geq 2$, Jensen's inequality, monotonicity of h , and $h(2) = 8$ give

$$\prod_{i=0}^{q-2} \rho_i^{-1} \leq h\left(\frac{q}{q-1}\right)^{q-1} \leq 8^{q-1}. \quad (67)$$

Thus (67) holds for every $q \geq 1$.

For the terminal factor,

$$\omega_{q-1} \leq M_q, \quad \omega_{q-1} + \Omega_q = \omega_{q-1} + 2\omega_{\text{end}} - 1 \leq 2(\omega_{q-1} + \omega_{\text{end}}) \leq 2M_q,$$

and hence

$$\begin{aligned} \Omega_q \rho_{q-1}^{-1} &= \omega_{q-1} \left(1 + \frac{\omega_{q-1} + \Omega_q}{\widehat{\delta}_q} \right) \\ &\leq M_q \left(1 + \frac{2M_q}{\delta_{\min}} \right) \leq 3qM_q \leq 3q_a M_{q_a}. \end{aligned} \quad (68)$$

The last inequality follows from (65) and $q(q_a + 1)/(q + 1) \leq q_a$. Combining (67) and (68), the selected prefix has certificate contribution at least

$$\frac{1}{3M_{q_a} q_a 8^{q-1}} \geq \frac{1}{4M_{q_a} q_a 8^{q_a-1}}.$$

□

The following internally derived corollary corresponds to the $q_\star = 3$ bounded-rank consequence in Ma–Chen [13, Lemma 6.5].

Corollary 6.2 (Uniform finite-cutoff constant). *Under the schedule and cutoff notation of Lemma 6.1, set $q_\star = 3$ and*

$$c_{\text{fin}} := \frac{1}{4q_\star 8^{q_\star-1}} = \frac{1}{768}. \quad (69)$$

Then, for every $0 \leq q_a \leq \min\{q_\star, N\}$,

$$\mathcal{H}_T(\alpha) \geq \frac{c_{\text{fin}}}{M_{q_a}}. \quad (70)$$

Proof. This follows from Lemma 6.1, since $q_a 8^{q_a-1} \leq q_\star 8^{q_\star-1}$ for $1 \leq q_a \leq q_\star$, while $c_{\text{fin}} \leq 1/2$ handles $q_a = 0$. □

Proposition 6.3 (Certificate lower bound). *There is a universal $c_{\text{sched}} > 0$ such that, for every horizon $T \geq 1$ and every dimensionless nonnegative schedule $\alpha \in \mathbb{R}_{\geq 0}^T$,*

$$\mathcal{H}_T(\alpha) \geq \frac{c_{\text{sched}}}{M_{\text{cap}}(N+1)^{\sqrt{3}-1}}. \quad (71)$$

One explicit choice is

$$c_{\text{sched}} := \min \left\{ c_{\text{fin}}, \frac{q_{\star}^{\beta_{\star}-1}}{\Gamma_{\text{prop}}} \min \left\{ \frac{1}{5}, c_{\text{fin}} \right\} \right\}. \quad (72)$$

Proof. The organization parallels the Ma–Chen cutoff scan [13, Section 6.3], with Corollary 4.3 replacing their matching-based favorable cutoff certificate. Every result invoked below has already been proved in this paper. Scan $q = q_{\star}, \dots, N$ and consider

$$\mathcal{Q}_{\text{low}} := \{q \in \{q_{\star}, \dots, N\} : d_q \leq d_{\star}\}.$$

If $\mathcal{Q}_{\text{low}} \neq \emptyset$, let $k = \max \mathcal{Q}_{\text{low}}$. Corollary 4.3 gives $\mathcal{H}_T(\alpha) \geq 1/(5M_k)$. By maximality, $d_q > d_{\star}$ for $k < q \leq N$, so Lemma 5.3 gives

$$M_k k^{\beta_{\star}-1} \leq \Gamma_{\text{prop}} M_{\text{cap}} N^{\beta_{\star}-1}.$$

Since $k \geq q_{\star}$,

$$\mathcal{H}_T(\alpha) \geq \frac{q_{\star}^{\beta_{\star}-1}}{5\Gamma_{\text{prop}} M_{\text{cap}} (N+1)^{\beta_{\star}-1}}.$$

Suppose $\mathcal{Q}_{\text{low}} = \emptyset$. If $N \geq q_{\star}$, apply Lemma 5.3 at $k = q_{\star}$, and then Corollary 6.2 at $q_a = q_{\star}$:

$$\mathcal{H}_T(\alpha) \geq \frac{c_{\text{fin}}}{M_{q_{\star}}} \geq \frac{c_{\text{fin}} q_{\star}^{\beta_{\star}-1}}{\Gamma_{\text{prop}} M_{\text{cap}} (N+1)^{\beta_{\star}-1}}.$$

If $N < q_{\star}$, take $q_a = N$. Since $M_N = M_{\text{cap}}$,

$$\mathcal{H}_T(\alpha) \geq \frac{c_{\text{fin}}}{M_{\text{cap}}} \geq \frac{c_{\text{sched}}}{M_{\text{cap}} (N+1)^{\beta_{\star}-1}}.$$

The definition (72) covers all three cases. □

Combining Proposition 6.3 with (8) gives the normalized horizon bound

$$\mathcal{H}_T(\alpha) \geq c_{\text{sched}} (T+1)^{-\sqrt{3}}. \quad (73)$$

Indeed,

$$M_{\text{cap}} (N+1)^{\sqrt{3}-1} \leq (T+1)^{1+\sqrt{3}-1} = (T+1)^{\sqrt{3}}.$$

7 Scaling, prescribed initialization, and proof of the theorem

The remaining normalization, translation, and scaling are standard; the details are included to keep the prescribed-initialization quantifiers explicit.

By (18) and (73), there is a convex 1-smooth Φ , a normalized initial point $\hat{x}_0$ with $\|\hat{x}_0\| = 1$, and a minimizer at 0, such that

$$\Phi(\hat{x}_T) - \Phi(0) \geq \frac{c_{\text{sched}}}{2} (T+1)^{-\sqrt{3}}. \quad (74)$$

Its dimension is $J+1 \leq N+1 \leq T+1$.

Return to the physical schedule η_t by setting $\alpha_t = L\eta_t$. Given any prescribed $x_{\text{init}} \in \mathbb{R}^d$, define

$$x_{\star} := x_{\text{init}} - R\hat{x}_0, \quad f(x) := LR^2\Phi\left(\frac{x - x_{\star}}{R}\right). \quad (75)$$

Then

$$\nabla f(x) = LR \nabla \Phi \left(\frac{x - x_\star}{R} \right),$$

so f is convex and L -smooth, $x_\star \in \operatorname{argmin} f$, and $\|x_{\text{init}} - x_\star\| = R$. If $x_t = x_\star + R\hat{x}_t$, then

$$\begin{aligned} x_{t+1} &= x_\star + R\hat{x}_t - \eta_t LR \nabla \Phi(\hat{x}_t) \\ &= x_\star + R\{\hat{x}_t - \alpha_t \nabla \Phi(\hat{x}_t)\} = x_\star + R\hat{x}_{t+1}. \end{aligned}$$

Thus the physical iterates reproduce the normalized trajectory. Finally,

$$\begin{aligned} f(x_T) - f(x_\star) &= LR^2 \{\Phi(\hat{x}_T) - \Phi(0)\} \\ &\geq \frac{c_{\text{sched}}}{2} LR^2 (T+1)^{-\sqrt{3}}. \end{aligned}$$

Theorem 1.2 follows with

$$c_{\text{lb}} := \frac{c_{\text{sched}}}{2} > 0. \quad (76)$$

For the explicit choices in (46) and (54), we have $c_{\text{fin}} < 1/5$ and $\Gamma_{\text{prop}} > 3 > 3^{\sqrt{3}-1} = q_\star^{\beta_\star-1}$. Thus the second term realizes the minimum in (72). Substitution into $c_{\text{lb}} = c_{\text{sched}}/2$ gives

$$c_{\text{lb}} = \frac{3^{\sqrt{3}-1}}{3072 \exp((2+3\pi^2)/(1+\sqrt{3}))}. \quad (77)$$

8 Discussion and limitations

Theorem 1.2 establishes a universal last-iterate obstruction for plain GD with predetermined nonnegative stepsizes: regardless of the announced schedule, a convex L -smooth instance in dimension at most $T+1$ has error at least a constant multiple of $LR^2(T+1)^{-\sqrt{3}}$. The conclusion holds for every prescribed initialization, and the constant is independent of the horizon, schedule, dimension, and scale parameters.

The central mechanism is the exact temporal inverse product. Keeping the shared terminal correction inside the full unrelaxed product permits a two-state entropy estimate to telescope over the selected path. All interior entropy terms cancel, and the remaining endpoint expression is bounded uniformly. The resulting critical-density implication at $1/2$, combined with the endpoint Lyapunov drift and the finite-cutoff scan, yields the exponent $\sqrt{3}$. The endpoint estimates and the final numerical constant are deliberately unoptimized.

The proof should be read as a refinement of the Ma–Chen framework [13], not as a separate proof architecture. Its marked-stage realization, residual-rank dynamics, Lyapunov tail propagation, bounded-rank prefix estimate, and final cutoff scan have counterparts in their paper, but each is proved in full within this manuscript. Thus Ma–Chen is an attributional comparison, not a logical dependency. The specific contribution is the direct endpoint-entropy control of the full ordered product built from the shared exact local and terminal identities. It replaces their budget-equalization and matching analysis and strengthens the favorable-cutoff certificate sufficiently to reach exponent $\sqrt{3}$; it does not assert that the resulting exponent is optimal.

The silver-schedule constructions leave the nontrivial bracket

$$\log_2(1 + \sqrt{2}) \leq \beta_{\text{GD}} \leq \sqrt{3}.$$

Closing this gap remains open. In particular, the present result does not claim that the density threshold $1/2$, the factor $1/5$, or the exponent $\sqrt{3}$ is optimal.

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